

Equations & inequalities

Understand the question → choose the tool → explain the answer

01 What is this topic about?

An **equation** asks for values that make two expressions equal. An **inequality** asks for a range of values. A **root** is a value of x that makes the quadratic equal to zero.

THE TWO TOOLS YOU NEED

$$ax^2 + bx + c = 0, \quad a \neq 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$D = b^2 - 4ac$$

D is the **discriminant**: it tells you how many real roots exist.

a , b and c include their signs. Rearrange to zero before identifying them.

02 When do I use each one?

Need the values of x ? Factorise if easy; otherwise use the quadratic formula. **Only need the nature of roots?** Calculate D , not the roots themselves.

$$D > 0$$

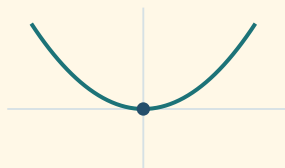
Two distinct real roots



Crosses the x-axis twice.

$$D = 0$$

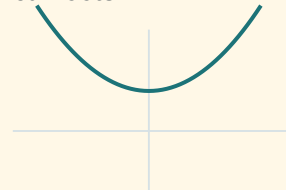
Two equal real roots



One repeated root; touches.

$$D < 0$$

No real roots



Does not meet the x-axis.

03 Try the formula once

$$x^2 + 2x - 2 = 0 \Rightarrow a = 1, b = 2, c = -2$$

$$x = \frac{-2 \pm \sqrt{2^2 - 4(1)(-2)}}{2(1)} = \frac{-2 \pm \sqrt{12}}{2} = -1 \pm \sqrt{3}$$

The $\pm$ gives two answers. Leave them exact when asked, rather than rounding early.

Equations & inequalities

Discriminants / exam wording / unknown constants

04 Turn the wording into a condition

A parameter, such as k or m , is an unknown constant, not the variable x .

WORDS IN THE QUESTION

Distinct roots / two different intersections

Equal / repeated roots / tangent / touches

No real roots / no intersection

Real roots / at least one intersection

Positive for every real x

Negative for every real x

CONDITION

$$D > 0$$

$$D = 0$$

$$D < 0$$

$$D \geq 0$$

$$a > 0 \quad \text{and} \quad D < 0$$

$$a < 0 \quad \text{and} \quad D < 0$$

05 Worked example: a range of constants

Find k if the equation below has real roots.

$$x^2 + kx + 9 = 0$$

Use $D \geq 0$ because a repeated root is allowed.

$$k^2 - 4(1)(9) \geq 0 \implies (k - 6)(k + 6) \geq 0$$

$$k \leq -6 \quad \text{or} \quad k \geq 6$$

For distinct roots, exclude ± 6 . For equal roots, $k = -6$ or $k = 6$.

06 Worked example: always negative

Find m if the expression below is strictly negative for every real x .

$$-x^2 + 4x + m$$

$a = -1 < 0$ already. Also require $D < 0$:

$$4^2 - 4(-1)m < 0 \implies 16 + 4m < 0 \implies m < -4$$

Exam tip: $D < 0$ alone does not tell you whether the whole graph is above or below the x -axis. Check the sign of a . At $m = -4$ it touches zero, so exclude it.

Equations & inequalities

Simultaneous equations / intersections / tangency

07 Solve two equations at the same time

A solution must satisfy **both** equations. For one linear and one non-linear equation, substitution turns the problem into one equation in one variable.

EXAMPLE: FIND BOTH INTERSECTION POINTS

$$y = 2x + 1, \quad y = x^2 - 2x + 4$$

1. Substitute the linear expression for y into the curve.

$$2x + 1 = x^2 - 2x + 4$$

$$x^2 - 4x + 3 = 0 \Rightarrow (x - 1)(x - 3) = 0$$

2. Solve for x, then substitute EACH x-value back into the line.

$$x = 1 \Rightarrow y = 3; \quad x = 3 \Rightarrow y = 7$$

$$(x, y) = (1, 3) \text{ or } (3, 7)$$

Check: substitute each pair into both originals. If the other equation contains y^2 , substitute the **whole bracket** before squaring, not just the x term.

08 Only need the number of intersections?

First equate the line and curve and rearrange to zero. Use the discriminant of **that new quadratic**, not automatically the original curve.

EXAMPLE: FIND t WHEN THE LINE IS A TANGENT

$$y = x + t, \quad y = x^2 - 3x + 1$$

$$x^2 - 4x + 1 - t = 0$$

$$D = (-4)^2 - 4(1)(1 - t) = 12 + 4t$$

$$D = 0 \Rightarrow t = -3$$

Then $(x - 2)^2 = 0$ gives the single contact point (2, -1).

Equations & inequalities

Quadratic inequalities / sign intervals / number lines

09 Find a range, not just the roots

Method: move everything to one side → find the boundary roots → check the sign in each interval → choose the required intervals → draw the number line.

EXAMPLE: AT OR BELOW ZERO

$$x^2 - 5x + 6 \leq 0 \Rightarrow (x - 2)(x - 3) \leq 0$$

Boundary roots: 2 and 3. Since $a > 0$, the graph opens upwards. It is negative between the roots and positive outside them.

$$x < 2$$

+

$$2 < x < 3$$

-

$$x > 3$$

+

$$2 \leq x \leq 3$$

Include the roots because equality is allowed.



10 What if the x^2 coefficient is negative?

$$-x^2 + x + 6 > 0 \Rightarrow -(x - 3)(x + 2) > 0$$

The roots are -2 and 3. The graph opens downwards, so it is **positive between** them. Therefore $-2 < x < 3$. Use hollow dots: equality is not allowed.



EXAM TRAPS TO AVOID

Do not lose a branch: $x^2 > 36$ means $x < -6$ **or** $x > 6$, not just $x > 6$.

Reverse the sign if you multiply/divide by a negative number. Do not divide by a variable whose sign is unknown.

Do not memorise “inside” blindly: check a or test one value in each interval.

Special roots: $(x - 2)^2 < 0$ has no solution; $(x - 2)^2 \leq 0$ gives only $x = 2$.

Answer format: use “or” for two separate ranges; filled dots for $\leq/\geq$, hollow dots for $</>$.

Practice: equations & inequalities

Practice 1 of 3 / Try without the notes first

Name: _____ Date: _____

01 Solve exactly

[3 marks]

Solve the equation. Give both answers in exact form.

$$2x^2 + 3x - 7 = 0$$

Original exam-style question

02 Interpret the discriminant

[5 marks]

For the equation below, find:

- (a) the values of k giving two equal roots;
- (b) the range of k giving real roots (including equal roots).

$$2x^2 + kx + 8 = 0$$

Original exam-style question

Practice: equations & inequalities

Practice 2 of 3 / Keep both solutions and show the interval

03 Simultaneous equations

[5 marks]

Solve the simultaneous equations.

$$y = x + 1, \quad x^2 + y^2 = 13$$

Original exam-style question

04 Quadratic inequality

[4 marks]

Solve the inequality and show your answer on a number line.

$$2x^2 + x - 6 \leq 0$$

Original exam-style question

Practice: equations & inequalities

Practice 3 of 3 / Apply conditions to unknown constants

05 Always positive

[3 marks]

Find all p such that the expression stays strictly positive for every real x .

$$x^2 - (p + 3)x + 2p + 3$$

[2024, Jurong West Secondary School, WA1 Q2]

Wording shortened; expression unchanged.

06 A line and a curve

[7 marks]

Find the range or value of k for which the line meets the curve:

(a) at two distinct points; (b) as a tangent; (c) at no point.

For (b), also find the coordinates of the point of contact.

$$y = 2x + k, \quad y = x^2 - 4x + 7$$

Original exam-style question

Worked answers

Self-check 1 of 2 / Compare your method, not only the final answer

01 Quadratic formula

$$a = 2, \quad b = 3, \quad c = -7$$

$$x = \frac{-3 \pm \sqrt{3^2 - 4(2)(-7)}}{2(2)} = \frac{-3 \pm \sqrt{65}}{4}$$

Check: both signs are needed. The discriminant is $65 > 0$, so there must be two distinct real roots.

02 A parameter in the equation

$$D = k^2 - 4(2)(8) = k^2 - 64$$

(a) Equal roots require $D = 0$:

$$k^2 - 64 = 0 \Rightarrow k = -8 \quad \text{or} \quad k = 8$$

(b) Real roots require $D \geq 0$:

$$(k - 8)(k + 8) \geq 0 \Rightarrow k \leq -8 \quad \text{or} \quad k \geq 8$$

Include the endpoints in (b): “real” does not mean “distinct”.

03 Substitute the whole expression

$$x^2 + (x + 1)^2 = 13$$

$$2x^2 + 2x - 12 = 0 \Rightarrow x^2 + x - 6 = 0$$

$$(x + 3)(x - 2) = 0 \Rightarrow x = -3 \quad \text{or} \quad x = 2$$

Substitute into $y = x + 1$:

$$(x, y) = (-3, -2) \quad \text{or} \quad (2, 3)$$

Check both originals: $(-3)^2 + (-2)^2 = 13$ and $2^2 + 3^2 = 13$; each pair also satisfies $y = x + 1$. Do not combine one x-value with the wrong y-value.

Worked answers

Self-check 2 of 2 / Signs, endpoints and interpretation

04 Choose the correct interval

$$(2x - 3)(x + 2) \leq 0$$

$$x = -2 \quad \text{or} \quad x = \frac{3}{2}$$

The upward-opening quadratic is at or below zero between its roots. Include both endpoints.

$$-2 \leq x \leq \frac{3}{2}$$



05 School question: always positive

The leading coefficient is $1 > 0$. For strict positivity, also require $D < 0$.

$$D = [-(p + 3)]^2 - 4(1)(2p + 3) = p^2 - 2p - 3$$

$$(p - 3)(p + 1) < 0 \implies -1 < p < 3$$

Exclude -1 and 3: at either endpoint the graph touches the x-axis, so it is not strictly positive for every x.

06 Use the quadratic formed at the intersection

$$2x + k = x^2 - 4x + 7 \implies x^2 - 6x + 7 - k = 0$$

$$D = (-6)^2 - 4(1)(7 - k) = 8 + 4k$$

(a) Two distinct points: $D > 0$ gives $k > -2$.

(b) Tangent: $D = 0$ gives $k = -2$.

$$x^2 - 6x + 9 = 0 \implies (x - 3)^2 = 0$$

$x = 3$; $y = 2(3) - 2 = 4$. Point of contact: (3, 4).

(c) No intersection: $D < 0$ gives $k < -2$.

SCOPE & QUESTION SOURCES

Notes rebuilt from Elyssa's S3A Flash Recap (2025), Chapter 2, with corrected qualifications. Scope checked against SEAB K341 (2027), A2; always-positive/negative conditions connect to A1. Questions 1-4 and 6 are original exam-style practice, not copied school questions. Their marks are suggested. Question 5: Jurong West Secondary, 2024 WA1, Q2 (3 marks); wording shortened, algebra unchanged. [SEAB syllabus](#) | School-paper source. School chapter sequencing may vary.